

Structured Output and Schemas

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Iterative Prompting

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompt as a Contract

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

A Distribution of Possibilities

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Audit types and fields for Structured outputs and validation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Structured outputs and validation. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Validate new objects independently

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required fields. Y replaces 9 with "9". Z keeps integer 9 but names room G. Trace the first failing layer for each and explain the difference.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Schema requires title:text and pages:number. Output is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

6. Solve the given case

Schema requires title text and pages number. Output is {"title":"River","pages":"80"}.

Diagnose the shape error.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A required schema has title as text and count as an integer. The output is {"title":"Seeds","count":"three"}. Identify the error and give a valid replacement.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 8 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Require answer, source snippet, citation, uncertainty, and next-question fields.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Schema requires title:text and pages:number. Output is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Distinguish the layers

Explain what parsing, schema checks and source support each establish.

My reasoning and evidence

13. W2 Audit types and fields

Why do string "12",boolean true,missing capacity and an extra action field violate this contract?

My reasoning and evidence

14. W3 Check a citation-like ID

Sources P,Q support B capacity 12. A schema-valid object cites P,Q but gives 99. Which layer rejects it?

My reasoning and evidence

15. W4 Validate new objects independently

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required fields. Y replaces 9 with "9". Z keeps integer 9 but names room G. Trace the first failing layer for each and explain the difference.

My reasoning and evidence

16. W5 Keep permissions separate

A fully valid supported reporting object is returned. May the application automatically book the room? Explain.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Structured Output and Schemas

A fictional Study Buddy returns a machine-readable book fact.

Schema: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical fields.

Structured Output and Schemas

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Parse

Each shown object is valid JSON syntax.



2. Schema check

Y fails because its pages value is text, not an integer.



3. Meaning check

Z fits field types but contradicts R's page count.



4. Accept X

X passes syntax, declared schema and source comparison.

CHANGE ONE THING

Add an extra field "rating":5 to X under the exactly-fields schema.

Explain the changed result using the matching lesson.

Which object demonstrates that valid JSON and valid types are insufficient for truth?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Validate Study Buddy outputs before another program consumes them; retain source IDs and separate syntax/type/support failures.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Validate Study Buddy outputs before another program consumes them; retain source IDs and separate syntax/type/support failures.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy returns a machine-readable book fact.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? X passes syntax, declared schema and source comparison.

1. Meaning check
2. Accept X
3. Schema check
4. Parse

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Accept X X passes syntax, declared schema and source comparison. Parsing establishes that text forms a structured object. Schema checks establish required fields and types. Semantic validation compares their meaning with the approved source. These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

Practice 2: explain the deciding evidence

Changed-case practice: Add an extra field "rating":5 to X under the exactly-fields schema. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The changed object fails schema despite keeping the correct page count.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The changed object fails schema despite keeping the correct page count. Exactly means additional fields are not allowed in this contract.

Repair a missing building block

Review Iterative Prompting (ai-3/7). Goal: Explain the mechanism and terminology of controlled prompt experiments. Short practice: What evidence supports improvement here, and what remains unproven?. Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?. Return to this week and explain what you changed.

Review A Distribution of Possibilities (ai-3/2). Goal: Check whether a next-token table is a valid probability distribution. Short practice: Why is 0.625 not a broken probability sum?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
